

DES Engine

Reference Manual

A discrete-event simulation engine in the spirit of Arena's Basic Process template, written from the simulation theory table up. This manual lists every public type and member, says what each one is for, and traces what actually calls what when a model runs.

The signatures here are **extracted from the headers** when the manual is built, so they cannot drift from the code. Only the prose is written by hand.

v10 -- the expression layer - 543 checks - 86 types - 578 public members

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Chapter 1

Reading This Manual

What is in here, how it was produced, and the shape of the engine as a whole.

This manual documents every public type and member of the engine. It is generated: **the signatures are read out of include/*.hpp when the PDF is built**, so they cannot drift from the code. Only the prose is written by hand. If a member appears here it exists; if one is deleted from the code it disappears from the manual on the next build.

Conventions

Shown	Means
class	Has invariants to protect; members default to private.
struct	A bag of values, all public.
enum class	A closed set of named alternatives.
inherits	Base classes, so the polymorphism is visible.
I-prefix	An interface: pure virtual, implemented elsewhere.

Members are listed in header order rather than alphabetically, because header order is how the author grouped them and that grouping carries meaning.

The engine in layers

Experiments Experiment - Summary - replications, intervals
Engine SimulationSystem - Clock - FutureEventList - Trace
Model Model - INode - Station - Resource
Expressions Parser - IExpression - EvalContext - VariableStore
Foundations Entity - Statistics - RandomStream - Diagnostic

Each layer uses the ones below it. The expression layer sits at the bottom rather than beside the model, because it must not depend on Model: it declares IModelState and the engine implements it.

The one rule worth knowing first

Inside the run loop the integrals for the interval that just ended are closed **before** the clock moves, and the clock moves **before** any state changes. Swap any two of those and every time-weighted average goes quietly wrong, with no error anywhere. Chapter 10 shows the order explicitly.

Chapter 2

Foundations

Time, entities, events, and the accumulators every reported number comes from.

EventType

enum class · Common.hpp

What kind of thing an event notice represents. The engine dispatches on this in the run loop.

Values Arrival, StartService, Departure, EndSimulation, WarmUpEnd, Observe, Renege

QueueDiscipline

enum class · Common.hpp

The service order a queue uses. Selects which IQueueRule a station is built with.

Values FIFO, LIFO, Priority, Random, SPT, EDD

ResourceState

enum class · Common.hpp

Whether a resource is idle, partly busy, or fully seized.

Values Idle, Busy

Clock

class · Clock.hpp

Simulation time, and the invariant that it never goes backwards. `advanceTo()` is the only door, which is why the class exists rather than a bare double.

SimTime now() const
The current simulation time.

void advanceTo(SimTime t)
Moves time forward. Refuses to go backwards -- the invariant the class exists to protect.

void reset()
Back to the `t=0` condition. Configuration survives; run state does not.

Entity

class · Entity.hpp

The thing that flows through the model: a customer, a part, a patient. Carries an id, a creation time, a type name, and an open-ended bag of numeric attributes.

EntityId id() const
The unique id assigned at creation.

SimTime creationTime() const
When this entity entered the system. Copies inherit the original's, because they are the same work.

void setAttribute(const std::string& name, double value)
Inserts or overwrites an attribute.

double attribute(const std::string& name) const
Reads a numeric attribute. Returns 0.0 when absent, which is why callers that care check `hasAttribute()` first.

bool hasAttribute(const std::string& name) const
Whether the attribute has ever been set on this entity.

void copyAttributesFrom(const Entity& other)
Copies every attribute from another entity. Used when duplicating.

void addMember(Entity* e)
Adds an entity to this one's batch. A batch representative carries its members until a Separate splits them out.

const std::vector<Entity*> members() const
The entities this batch representative is carrying.

bool isBatch() const

Whether this entity is carrying members.

void clearMembers()

Drops the carried members, after a Separate has released them.

void setCreationTime(SimTime t)

Overrides the creation time. Used when a copy must inherit the original's, because it is the same work.

const std::string& type() const

The entity's type name, which per-type reporting is keyed on.

void setType(std::string t)

Changes the type name. Prefer `NodeContext::setEntityType`, which also moves the live per-type count.

EventNotice

class · EventNotice.hpp

One scheduled future occurrence: a time, a kind, and optionally the entity and block it concerns. Ordered by time, with a sequence number breaking ties so equal-time events keep their scheduling order.

EventType type() const
The type.

SimTime time() const
The time.

Entity* entity() const
The entity.

Inode* node() const
The node.

uint64_t sequenceNumber() const
A monotonically increasing stamp used only for tie-breaking.

bool operator>(const EventNotice& other) const
Orders notices for the heap: earlier times first, with the sequence number breaking ties so equal-time events keep scheduling order.

FutureEventList

class · FutureEventList.hpp

The pending-event set, kept as a binary heap ordered by time. `popImminent()` is the beating heart of the run loop. It cannot remove an arbitrary element, which is why reneging uses lazy cancellation instead.

bool isEmpty() const
Whether anything is still scheduled. The run loop stops when this is true.

std::size_t size() const
How many events are pending.

void clear()
Discards every pending event. Part of resetting between replications.

void schedule(EventNotice e)
Inserts a notice into the heap.

EventNotice popImminent()
Removes and returns the earliest notice. The centre of the run loop.

SimTime nextEventTime() const
When the next event is due, without removing it.

Activity

class · Activity.hpp

A span whose duration is known when it starts -- a service time drawn at seize. `endTime()` is derived, never stored, so the two cannot disagree.

Activity(std::string name, SimTime start, SimTime duration)
 Constructor.

const std::string& name() const
 The name this object was constructed with.

SimTime startTime() const
 The start time.

SimTime duration() const
 The duration.

SimTime endTime() const
 Start plus duration, derived rather than stored so the two cannot disagree.

Delay

class · Delay.hpp

A span whose duration is decided later, by the system rather than by a draw -- how long an entity waited for a server to free up.

explicit Delay(SimTime start)
 Constructor.

SimTime startTime() const
 The start time.

bool hasEnded() const
 Whether this delay has been closed.

void end(SimTime t)
 Closes the delay at the given time.

SimTime duration() const
 The duration.

SystemState

class · SystemState.hpp

A snapshot of the whole system: number in system, number queued, server status. Summed across stations rather than owned by any one of them.

int numberInSystem() const
 The number in system.

int numberInQueue() const
 The number in queue.

ResourceState serverStatus() const
 Whether servers are idle, busy or fully seized.

void setNumberInSystem(int count)
 Sets the number in system.

void setNumberInQueue(int count)
 Sets the number in queue.

void setServerStatus(ResourceState status)
 Sets the server status.

void reset()
 Back to the t=0 condition. Configuration survives; run state does not.

Statistics

class · Statistics.hpp

The accumulators behind every reported average. Holds counts, totals and two time integrals, and knows the difference between a tally average and a time-weighted one.

Statistics(std::string labelA, std::string labelB)
 Constructor.

int numberArrived() const
 How many arrivals were counted.

int numberServed() const
 How many departures were counted.

SimTime totalWaitingTime() const
 The total waiting time.

SimTime totalTimeInSystem() const
 The total time in system.

SimTime maxWaitingTime() const
 The max waiting time.

SimTime areaA() const
 The raw first integral. Divide by measured time for the average.

SimTime areaB() const
 The raw second integral. Divide by measured time for the average.

SimTime lastUpdateTime() const
 When the integrals were last closed.

const std::string& labelA() const
 The label a.

const std::string& labelB() const
 The label b.

void recordArrival(SimTime t)
 Counts an arrival.

void recordDeparture(SimTime t, SimTime waitTime, SimTime timeInSystem)
 Counts a departure and accumulates its waiting time and time in system.

void updateTimeIntegrals(SimTime now, int valueA, int valueB)
 Closes the interval that just ENDED, so it must be called before the clock moves and before any state changes, with the OLD values.

void reset()
 Back to the t=0 condition. Configuration survives; run state does not.

void restartAt(SimTime now)
 Throws away everything collected and starts accumulating from now. Warm-up removal in one line.

double averageWaitingTime() const
 The average waiting time.

double averageTimeInSystem() const
 The average time in system.

double timeAverageA(SimTime elapsed) const
 The first time-weighted average over the measured interval. The caller supplies the elapsed time so that warm-up handling is visible.

double timeAverageB(SimTime elapsed) const
 The second time-weighted average over the measured interval.

double utilisation(SimTime elapsed, int capacity) const
 Busy server-time divided by capacity times elapsed time.

ModelError

class · ModelError.hpp · inherits public std::runtime_error

Thrown when a model is wired wrongly -- a programmer's mistake, where stopping is the right response. Contrast Diagnostic, which is how a user's mistake is reported.

explicit ModelError(const std::string& what) :
std::runtime_error(what)
 Constructor.

Chapter 3

Queues and Resources

Waiting lines, the disciplines that order them, and the servers they wait for.

A queue holds entities; a rule decides which one leaves next; a resource is the pool of servers they are waiting for. Resources are owned by the **Model** rather than by a block, which is what makes one operator shared across three machines expressible at all.

EntityQueue

class · EntityQueue.hpp

A waiting line: a deque of entities plus a pluggable rule that chooses which one leaves next. Tracks its own maximum observed length.

```
EntityQueue(const std::string& name, QueueDiscipline
discipline)
    Constructor.

EntityQueue(const std::string& name,
std::unique_ptr<IQueueRule> rule)
    Constructor.

const std::string& name() const
    The name this object was constructed with.

std::string ruleName() const
    The discipline's name, for reports.

std::size_t length() const
    The length.

bool isEmpty() const
    Whether anything is waiting here.

std::size_t maxLengthObserved() const
    The longest this queue has ever been.

const std::deque<Entity*>& contents() const
    The waiting entities, in arrival order, whichever discipline is in use.

void setRandomStream(RandomStream* rng)
    Supplies the stream the Random discipline draws from.

void reset()
    Back to the t=0 condition. Configuration survives; run state does not.

void resetStatistics()
    Discards measurements collected so far and starts again from now,
without touching what is held. This is warm-up removal at block level.

void push(Entity* e)
    Adds an entity to the back of the line.

Entity* pop()
    Removes and returns the entity the discipline selects.

bool remove(Entity* e)
    Removes a specific entity if it is still waiting, and reports whether it
was. How reneging works.
```

IQueueRule

class · QueueRule.hpp

The interface a queue discipline implements. One method: given the waiting entities, which index is served next.

```
virtual std::string name() const = 0
    The name this object was constructed with.
```

FifoRule

class · QueueRule.hpp · inherits public IQueueRule

First in, first out. Always index 0.

```
std::size_t selectIndex(const std::deque<Entity*>&,
RandomStream&) const override
    Which waiting entity is served next, as an index into the queue.

std::string name() const override
    The name this object was constructed with.
```

LifoRule

class · QueueRule.hpp · inherits public IQueueRule

Last in, first out. Always the final index.

```
std::size_t selectIndex(const std::deque<Entity*>&,
RandomStream&) const override
    Which waiting entity is served next, as an index into the queue.

std::string name() const override
    The name this object was constructed with.
```

HighestAttributeRule

class · QueueRule.hpp · inherits public IQueueRule

Serves whichever waiting entity has the largest value of a named attribute -- the Priority discipline. Ties resolve to the earliest arrival, deliberately.

```
explicit HighestAttributeRule(std::string attribute)
    Constructor.

std::size_t selectIndex(const std::deque<Entity*>&,
RandomStream&) const override
    Which waiting entity is served next, as an index into the queue.

std::string name() const override
    The name this object was constructed with.
```

LowestAttributeRule

class · QueueRule.hpp · inherits public IQueueRule

Serves the smallest value of a named attribute -- shortest processing time, or earliest due date.

```
explicit LowestAttributeRule(std::string attribute)
    Constructor.

std::size_t selectIndex(const std::deque<Entity*>&,
RandomStream&) const override
    Which waiting entity is served next, as an index into the queue.

std::string name() const override
    The name this object was constructed with.
```

RandomRule

class · QueueRule.hpp · inherits public IQueueRule

Picks a waiting entity uniformly at random, drawing from the queue's random stream.

```
std::size_t selectIndex(const std::deque<Entity*>&,
RandomStream&) const override
    Which waiting entity is served next, as an index into the queue.

std::string name() const override
    The name this object was constructed with.
```

IResourceUser

class · Resource.hpp

The interface a block implements to compete for a shared resource: can it start something, and how long has its longest-waiting entity been waiting.

```
virtual bool hasWaiting() const = 0
    Whether waiting.

virtual SimTime headOfLineSince() const = 0
    When this user's longest-waiting entity started waiting. The resource
compares this across every block sharing it, so the longest waiter
anywhere is served next.

virtual void startFromQueue(NodeContext& ctx) = 0
```

Asks this user to start serving its next queued entity, if it can. Returning without seizing tells the resource to stop offering.

Resource

class · Resource.hpp

A pool of interchangeable servers with a capacity. Owned by the Model, not by a block, which is what lets several blocks share one operator. When a unit frees, the resource offers it to whichever user has waited longest.

Resource(std::string name, int capacity)

Constructor.

const std::string& name() const

The name this object was constructed with.

int capacity() const

How many units this resource has.

int unitsBusy() const

The units busy.

int unitsAvailable() const

Capacity minus units busy, derived rather than stored.

bool isAvailable() const

Whether at least one unit is free.

ResourceState state() const

The state.

void reset()

Back to the t=0 condition. Configuration survives; run state does not.

void addUser(IResourceUser* user)

Registers a block as a competitor for this resource.

std::size_t userCount() const

How many blocks compete for this resource.

void offerFreedUnit(NodeContext& ctx)

Offers a freed unit to whichever registered user has the longest-waiting entity, so a shared resource does not favour the block that released it.

void seize(int units = 1)

Takes units. Asserts they are available; the caller must check first.

void release(int units = 1)

Returns units to the pool.

Chapter 4

Randomness

One primitive, twelve distributions, and the tests that show a generator is sound.

Everything random is built on a single `u01()` call, so a run is reproducible from one seed. Distributions may be given their own named streams, which is what lets you change a service time without the arrival pattern shifting underneath you.

EngineKind

enum class · RandomStream.hpp

Which underlying generator a `RandomStream` uses. Includes `RANDU`, kept precisely so its failure can be demonstrated.

Values MersenneTwister, LinearCongruential, Randu

RandomStream

class · RandomStream.hpp

The single source of randomness. Everything is built on one `u01()` primitive, so the whole engine is reproducible from one seed. Supports named substreams and antithetic variates.

```

unsigned seed() const
    The seed.

EngineKind engine() const
    The engine.

long long draws() const
    The draws.

void reset()
    Back to the t=0 condition. Configuration survives; run state does not.

double u01()
    The single primitive. Every distribution in the engine is built on this one call.

void setAntithetic(bool on)
    Makes the stream return 1-u instead of u, the basis of antithetic variates.

bool isAntithetic() const
    Whether antithetic.

RandomStream substream(const std::string& name) const
    A named, independent stream derived from the base seed. Naming is what keeps a stream stable as the rest of a model changes.

SimTime exponential(SimTime mean)
    One exponential sample with the given MEAN. The inversion is the only place the 1/lambda conversion happens.

SimTime uniform(SimTime low, SimTime high)
    One uniform sample on a closed interval.

SimTime normal(SimTime mean, SimTime stdDev)
    One normal sample, by the Box-Muller transform.

SimTime lognormal(SimTime logMean, SimTime logStdDev)
    One lognormal sample, from the mean and sd of the LOGARITHM.

SimTime weibull(SimTime scale, SimTime shape)
    One Weibull sample, by inversion.

SimTime erlang(SimTime meanOfEach, int phases)
    One Erlang sample: the sum of `phases' exponential draws.

SimTime triangular(SimTime low, SimTime mode, SimTime high)
    One triangular sample, by inversion either side of the mode.

int poisson(double mean)
    One Poisson count, by Knuth's multiplication method.

bool bernoulli(double p)
    True with probability p. One draw.

std::size_t uniformIndex(std::size_t n)
    A uniformly chosen index in [0, n). What the Random queue discipline uses.

static double normalQuantile(double p)
    The inverse normal CDF, used to build normal samples and confidence intervals.

```

IDistribution

class · Distribution.hpp

The interface every distribution implements: draw a value, report a theoretical mean, describe yourself, clone yourself, and optionally sample from your own stream.

```

virtual SimTime draw(RandomStream& rng) = 0
    Samples one value, using this object's own stream if it has been given one.

virtual std::string describe() const = 0
    A human-readable one-line description, used in model dumps and traces.

virtual void reset()
    Back to the t=0 condition. Configuration survives; run state does not.

virtual SimTime mean() const = 0
    The theoretical mean, used by the stability check before the run.

virtual std::unique_ptr<IDistribution> clone() const = 0
    A deep copy. Needed because a distribution may be held in two places at once.

void useStream(RandomStream* s)
    Gives every sampling site beneath this node its own random stream.

RandomStream* stream() const
    The stream.

```

Exponential

class · Distribution.hpp · inherits public IDistribution

Exponential inter-event times, parameterised by MEAN, not rate. The 1/lambda conversion happens in exactly one line.

```

explicit Exponential(SimTime mean)
    Constructor.

SimTime draw(RandomStream& rng) override
    Samples one value, using this object's own stream if it has been given one.

std::string describe() const override
    A human-readable one-line description, used in model dumps and traces.

SimTime mean() const override
    The theoretical mean, used by the stability check before the run.

std::unique_ptr<IDistribution> clone() const override
    A deep copy. Needed because a distribution may be held in two places at once.

```

Constant

class · Distribution.hpp · inherits public IDistribution

Always the same value. Consumes nothing from the random stream, which matters when comparing runs.

```

explicit Constant(SimTime value)
    Constructor.

SimTime draw(RandomStream& rng) override
    Samples one value, using this object's own stream if it has been given one.

std::string describe() const override
    A human-readable one-line description, used in model dumps and traces.

SimTime mean() const override
    The theoretical mean, used by the stability check before the run.

std::unique_ptr<IDistribution> clone() const override
    A deep copy. Needed because a distribution may be held in two places at once.

```


Uniform

class · Distribution.hpp · inherits public IDistribution

Uniform on a closed interval.

Uniform(SimTime low, SimTime high)
Constructor.

SimTime draw(RandomStream& rng) override
Samples one value, using this object's own stream if it has been given one.

std::string describe() const override
A human-readable one-line description, used in model dumps and traces.

SimTime mean() const override
The theoretical mean, used by the stability check before the run.

std::unique_ptr<IDistribution> clone() const override
A deep copy. Needed because a distribution may be held in two places at once.

Triangular

class · Distribution.hpp · inherits public IDistribution

Triangular with a low, a mode and a high. The usual choice when you have an expert's best, worst and likely case and no data.

Triangular(SimTime low, SimTime mode, SimTime high)
Constructor.

SimTime draw(RandomStream& rng) override
Samples one value, using this object's own stream if it has been given one.

std::string describe() const override
A human-readable one-line description, used in model dumps and traces.

SimTime mean() const override
The theoretical mean, used by the stability check before the run.

std::unique_ptr<IDistribution> clone() const override
A deep copy. Needed because a distribution may be held in two places at once.

Normal

class · Distribution.hpp · inherits public IDistribution

Normal, optionally truncated at zero so a duration cannot come out negative.

Normal(SimTime mean, SimTime stdDev, bool truncateAtZero = true)
Constructor.

SimTime draw(RandomStream& rng) override
Samples one value, using this object's own stream if it has been given one.

std::string describe() const override
A human-readable one-line description, used in model dumps and traces.

SimTime mean() const override
The theoretical mean, used by the stability check before the run.

std::unique_ptr<IDistribution> clone() const override
A deep copy. Needed because a distribution may be held in two places at once.

Lognormal

class · Distribution.hpp · inherits public IDistribution

Lognormal. Its constructor takes the mean and standard deviation of the LOGARITHM; use fromMeanAndSd() if you have the observed values instead.

Lognormal(SimTime logMean, SimTime logStdDev)
Constructor.

static std::unique_ptr<Lognormal> fromMeanAndSd(SimTime mean, SimTime sd)
Builds a Lognormal from the mean and standard deviation you observed, rather than those of the logarithm. The conversion is not

obvious and getting it wrong is silent.

SimTime draw(RandomStream& rng) override
Samples one value, using this object's own stream if it has been given one.

std::string describe() const override
A human-readable one-line description, used in model dumps and traces.

SimTime mean() const override
The theoretical mean, used by the stability check before the run.

std::unique_ptr<IDistribution> clone() const override
A deep copy. Needed because a distribution may be held in two places at once.

Weibull

class · Distribution.hpp · inherits public IDistribution

Weibull, by scale and shape. Common for time-to-failure.

Weibull(SimTime scale, SimTime shape)
Constructor.

SimTime draw(RandomStream& rng) override
Samples one value, using this object's own stream if it has been given one.

std::string describe() const override
A human-readable one-line description, used in model dumps and traces.

SimTime mean() const override
The theoretical mean, used by the stability check before the run.

std::unique_ptr<IDistribution> clone() const override
A deep copy. Needed because a distribution may be held in two places at once.

Erlang

class · Distribution.hpp · inherits public IDistribution

A sum of k exponential phases. fromMean() takes the mean you actually want.

Erlang(SimTime meanOfEachPhase, int phases)
Constructor.

static std::unique_ptr<Erlang> fromMean(SimTime totalMean, int phases)
Builds an Erlang from the total mean you want and a phase count.

SimTime draw(RandomStream& rng) override
Samples one value, using this object's own stream if it has been given one.

std::string describe() const override
A human-readable one-line description, used in model dumps and traces.

SimTime mean() const override
The theoretical mean, used by the stability check before the run.

std::unique_ptr<IDistribution> clone() const override
A deep copy. Needed because a distribution may be held in two places at once.

Discrete

class · Distribution.hpp · inherits public IDistribution

A finite set of values with probabilities. Takes INDIVIDUAL probabilities summing to one; Arena's DISC spelling takes cumulative ones and is converted at the grammar boundary.

Discrete(std::vector<SimTime> values, std::vector<double> probabilities)
Constructor.

SimTime draw(RandomStream& rng) override
Samples one value, using this object's own stream if it has been given one.

std::string describe() const override
A human-readable one-line description, used in model dumps and traces.

SimTime mean() const override
The theoretical mean, used by the stability check before the run.

std::unique_ptr<IDistribution> clone() const override
 A deep copy. Needed because a distribution may be held in two places at once.

Empirical

class · Distribution.hpp · inherits public IDistribution

Samples straight from observed data with linear interpolation between order statistics -- for when you have a hundred real service times and no idea which textbook curve produced them.

explicit Empirical(std::vector<SimTime> observations)
 Constructor.

SimTime draw(RandomStream& rng) override
 Samples one value, using this object's own stream if it has been given one.

std::string describe() const override
 A human-readable one-line description, used in model dumps and traces.

SimTime mean() const override
 The theoretical mean, used by the stability check before the run.

std::unique_ptr<IDistribution> clone() const override
 A deep copy. Needed because a distribution may be held in two places at once.

Poisson

class · Distribution.hpp · inherits public IDistribution

Counts rather than durations: arrivals per hour, defects per batch.

explicit Poisson(double mean)
 Constructor.

SimTime draw(RandomStream& rng) override
 Samples one value, using this object's own stream if it has been given one.

std::string describe() const override
 A human-readable one-line description, used in model dumps and traces.

SimTime mean() const override
 The theoretical mean, used by the stability check before the run.

std::unique_ptr<IDistribution> clone() const override
 A deep copy. Needed because a distribution may be held in two places at once.

Deterministic

class · Distribution.hpp · inherits public IDistribution

Walks a fixed list of values, optionally repeating. The basis of every hand-checkable test in this project, because it makes a run reproducible on paper.

explicit Deterministic(std::vector<SimTime> values, bool repeat = true)
 Constructor.

SimTime draw(RandomStream& rng) override
 Samples one value, using this object's own stream if it has been given one.

std::string describe() const override
 A human-readable one-line description, used in model dumps and traces.

SimTime mean() const override
 The theoretical mean, used by the stability check before the run.

std::unique_ptr<IDistribution> clone() const override
 A deep copy. Needed because a distribution may be held in two places at once.

void reset() override
 Back to the t=0 condition. Configuration survives; run state does not.

TestResult

struct · StreamTests.hpp

The outcome of one generator-quality test: a statistic, a threshold, and whether it passed.

StreamTests

class · StreamTests.hpp

Statistical tests a random stream should survive: chi-square uniformity, Kolmogorov-Smirnov, runs up and down, autocorrelation, and a 3-D serial test that exposes lattice structure.

static TestResult chiSquareUniformity(RandomStream& rng, int n, int bins = 10)

Bins many draws and compares the counts against a flat expectation. Catches a generator whose output is not evenly spread.

static TestResult kolmogorovSmirnov(RandomStream& rng, int n)
 Compares the empirical distribution against the uniform one by their largest gap. Sensitive where chi-square is not.

static TestResult runsUpAndDown(RandomStream& rng, int n)
 Counts ascending and descending runs against their expected number. Catches sequences that are uniform but not independent.

static TestResult autocorrelation(RandomStream& rng, int n, int lag = 1)
 Correlation between draws a given lag apart. A generator can pass every one-dimensional test and still fail this.

static TestResult serial3D(RandomStream& rng, int n, int binsPerAxis = 16)
 Bins consecutive triples in the unit cube. This is the test that exposes RANDU, whose triples fall on fifteen planes.

static std::vector<TestResult> runAll(RandomStream& rng, int n = 600000)
 Runs every test and collects the results.

static void report(const std::vector<TestResult>& results, const std::string& title)
 Prints a table of test results.

static double chiSquareCritical95(int degreesOfFreedom)
 The 95% critical value of chi-square, so a test can report pass or fail rather than a bare statistic.

Chapter 5

The Expression Layer

How a model's conditions, values and durations became text rather than compiled code.

Through v9 a service time was a constructed distribution and a condition was a lambda, so changing a model needed a compiler. v10 makes every one of those fields a string. There is no separate distribution field any more: EXPO(0.8) is a function in the grammar whose node owns one of the twelve IDistribution objects that already existed.

Value

A variant of double and string: what an expression evaluates to. Strings exist for one purpose, comparing an entity type. Booleans are numbers, 0 and 1, which is what lets a condition field and a duration field be the same kind of field.

Errors are data, not exceptions

A malformed expression is a person typing in a cell, not a programmer bug, so the lexer and parser never throw: they record a Diagnostic carrying a SourceSpan, recover, and keep going. One call reports many problems. The exception is `expr()`, which throws, because on the C++ API a bad expression really is a programmer error and there is no cell to point at.

ExpressionError

class · Value.hpp · inherits public ModelError

Thrown when an expression cannot do its job AT RUN TIME -- a string in arithmetic, division by zero, an attribute read with no entity.

```
explicit ExpressionError(const std::string& what) :
ModelError(what)
    Constructor.
```

SourceSpan

struct · Diagnostic.hpp

An offset and length within the text that was parsed. Carrying this from the lexer to the diagnostic is what lets a front end put a cursor on the exact bad character.

Severity

enum class · Diagnostic.hpp

Whether a diagnostic is an error or a warning.

Values Error, Warning

Diagnostic

struct · Diagnostic.hpp

A user's mistake, reported as DATA rather than thrown. The parser collects a list of these and keeps going, because a spreadsheet must show every bad cell at once and an exception can only carry one failure.

TokenKind

enum class · Lexer.hpp

The lexical categories of the expression grammar.

Values Number, Text, Identifier, Plus, Minus, Star, Slash, Percent, Caret, Bang, Less, LessEqual, Greater, GreaterEqual, EqualEqual, BangEqual, AndAnd, OrOr, LParen, RParen, Comma, End, Invalid

Token

struct · Lexer.hpp

One lexed token: its kind, its span, and its number or text.

LexResult

struct · Lexer.hpp

Everything one call to `tokenise()` produced: the tokens, always ending in End, and any diagnostics.

UnaryOp

enum class · Expression.hpp

The prefix operators: negation and logical not.

Values Negate, Not

BinaryOp

enum class · Expression.hpp

The infix operators, from arithmetic through comparison to the short-circuiting logicals.

Values Add, Subtract, Multiply, Divide, Modulo, Power, Less, LessEqual, Greater, GreaterEqual, Equal, NotEqual, And, Or

IExpression

class · Expression.hpp

A node in the parsed expression tree. Beyond `evaluate()`, the interesting members are `validate()`, which reports problems without throwing, and `meanIfKnown()`, which admits when a mean cannot be computed in advance.

```
explicit IExpression(SourceSpan span) : m_span(span)
    Constructor.
```

SourceSpan span() const

The range of source text this came from.

```
virtual Value evaluate(EvalContext& ctx) const = 0
    Computes this node's value in the given context.
```

```
virtual void validate(const ValidationContext& vc,
std::vector<Diagnostic>& out) const = 0
```

Appends every problem found. Never throws, so one call reports many.

```
virtual std::optional<double> meanIfKnown() const
```

The mean if it can be computed in advance, or nothing. Nothing means 'cannot tell', never 'zero'.

```
virtual std::string describe() const = 0
```

A human-readable one-line description, used in model dumps and traces.

```
virtual std::unique_ptr<IExpression> clone() const = 0
    A deep copy of the whole subtree.
```

```
virtual void useStream(RandomStream* s)
```

Gives every sampling site beneath this node its own random stream.

```
virtual void reset()
```

Back to the `t=0` condition between replications. Only a Deterministic distribution has anything to do, but an expression that owns one must pass it on.

LiteralExpression

class · Expression.hpp · inherits public IExpression

A number or a quoted string, known at parse time.

```
LiteralExpression(Value v, SourceSpan span) :
IExpression(span), m_value(std::move(v))
    Constructor.

const Value& value() const
    The value.

Value evaluate(EvalContext&) const override
    Computes this node's value in the given context.

void validate(const ValidationContext&,
std::vector<Diagnostic>&) const override
    Appends every problem found to the list. Does not throw and does not
    stop at the first.

std::optional<double> meanIfKnown() const override
    The mean if it can be computed in advance, or nothing. Nothing means
    'cannot tell', never 'zero'.

std::string describe() const override
    A human-readable one-line description, used in model dumps and
    traces.

ExpressionPtr clone() const override
    A deep copy. Needed because a distribution may be held in two places
    at once.
```

NameExpression

class · Expression.hpp · inherits public IExpression

A bare name, resolved LATE -- the parser stores only the text, and whether it is an attribute, a variable, Entity.Type or TNOW is decided at validation and evaluation. That is what stops the parser going stale as new kinds of name are added.

```
const std::string& name() const
    The text the parser captured, still unresolved.

Value evaluate(EvalContext& ctx) const override
    Computes this node's value in the given context.

void validate(const ValidationContext& vc,
std::vector<Diagnostic>& out) const override
    Appends every problem found to the list. Does not throw and does not
    stop at the first.

std::string describe() const override
    A human-readable one-line description, used in model dumps and
    traces.

ExpressionPtr clone() const override
    A deep copy. Needed because a distribution may be held in two places
    at once.
```

UnaryExpression

class · Expression.hpp · inherits public IExpression

A prefix operator applied to one operand.

```
Value evaluate(EvalContext& ctx) const override
    Computes this node's value in the given context.

void validate(const ValidationContext& vc,
std::vector<Diagnostic>& out) const override
    Appends every problem found to the list. Does not throw and does not
    stop at the first.

std::optional<double> meanIfKnown() const override
    The mean if it can be computed in advance, or nothing. Nothing means
    'cannot tell', never 'zero'.

std::string describe() const override
    A human-readable one-line description, used in model dumps and
    traces.

ExpressionPtr clone() const override
    A deep copy. Needed because a distribution may be held in two places
    at once.

void useStream(RandomStream* s) override
    Gives every sampling site beneath this node its own random stream.

void reset() override
    Back to the t=0 condition. Configuration survives; run state does not.
```

BinaryExpression

class · Expression.hpp · inherits public IExpression

An infix operator over two operands. The logical operators short-circuit, which is what makes a guarded division safe to write.

```
Value evaluate(EvalContext& ctx) const override
    Computes this node's value in the given context.

void validate(const ValidationContext& vc,
std::vector<Diagnostic>& out) const override
    Appends every problem found to the list. Does not throw and does not
    stop at the first.

std::string describe() const override
    A human-readable one-line description, used in model dumps and
    traces.

ExpressionPtr clone() const override
    A deep copy. Needed because a distribution may be held in two places
    at once.

void useStream(RandomStream* s) override
    Gives every sampling site beneath this node its own random stream.

void reset() override
    Back to the t=0 condition. Configuration survives; run state does not.
```

LambdaExpression

class · Expression.hpp · inherits public IExpression

Adapter wrapping a C++ predicate so the pre-v10 std::function API is a constructor for the one representation rather than a second code path.

```
Value evaluate(EvalContext& ctx) const override
    Computes this node's value in the given context.

void validate(const ValidationContext&,
std::vector<Diagnostic>&) const override
    Appends every problem found to the list. Does not throw and does not
    stop at the first.

std::string describe() const override
    A human-readable one-line description, used in model dumps and
    traces.

ExpressionPtr clone() const override
    A deep copy. Needed because a distribution may be held in two places
    at once.
```

DistributionExpression

class · Expression.hpp · inherits public IExpression

Adapter wrapping an IDistribution so the pre-v10 API keeps working. Also what a parsed EXPO(0.8) with constant arguments becomes.

```
Value evaluate(EvalContext& ctx) const override
    Computes this node's value in the given context.

void validate(const ValidationContext&,
std::vector<Diagnostic>&) const override
    Appends every problem found to the list. Does not throw and does not
    stop at the first.

std::optional<double> meanIfKnown() const override
    The mean if it can be computed in advance, or nothing. Nothing means
    'cannot tell', never 'zero'.

std::string describe() const override
    A human-readable one-line description, used in model dumps and
    traces.

ExpressionPtr clone() const override
    A deep copy. Needed because a distribution may be held in two places
    at once.

void useStream(RandomStream* s) override
    Gives every sampling site beneath this node its own random stream.

void reset() override
    Back to the t=0 condition. Configuration survives; run state does not.
```

ParseResult

struct · Parser.hpp

What parsing produced: a tree, and the diagnostics found on the way. The tree is never null; on error it holds a placeholder that is never evaluated.

IModelState

class · EvalContext.hpp

The four questions an expression may ask about a running system, plus the clock. Declared by the expression layer and implemented by SimulationSystem, so the expression layer never has to include Model.hpp.

```
virtual double queueLength(const std::string& blockName)
const = 0
    How many entities are waiting at a named block. Throws on an
    unknown name rather than reading as an empty queue.
virtual double resourceBusy(const std::string& name) const =
0
    Units of a named resource currently seized.
virtual double resourceCapacity(const std::string& name)
const = 0
    A named resource's capacity.
virtual double numberInSystem() const = 0
    How many entities are anywhere in the system.
virtual SimTime now() const = 0
    The current simulation time.
```

EvalContext

class · EvalContext.hpp

The only view of the engine an expression gets while a run is happening: the current entity, the variables, the model state and the random stream. Any of them may be absent, and each accessor says so rather than returning a plausible default.

```
const Entity* entity() const
    The entity.
VariableStore* variables() const
    The variables.
const IModelState* state() const
    The state.
RandomStream* rng() const
    The rng.
double attribute(const std::string& name) const
    Reads an entity attribute, throwing if there is no entity rather than
    returning zero.
const std::string& entityType() const
    The current entity's type name.
RandomStream& requireRng() const
    The random stream, or an error explaining that none was supplied.
const IModelState& requireState() const
    The live model state, or an error explaining that none was supplied.
```

FieldContext

enum class · EvalContext.hpp

Whether the field being validated has an entity to read from. A Create block's interarrival time is evaluated before its entity exists, so an attribute reference there is an error caught before the run.

Values HasEntity, NoEntity

ValidationContext

struct · EvalContext.hpp

What a name is allowed to resolve to, checked before the run: the declared variables and attributes, and whether an entity is available.

```
bool isVariable(const std::string& name) const
    Whether this name is a declared variable.
```

```
bool isAttribute(const std::string& name) const
    Whether this name is a declared attribute.
```

VariableStore

class · VariableStore.hpp

Arena's Variable data module. Global, numeric, time-persistent values that belong to the system rather than to any entity. A variable may not share a name with an attribute, and writing an undeclared one throws.

```
void declare(const std::string& name, double initialValue)
    Declares a variable with an initial value. Refuses a duplicate, and
    refuses a name already used by an attribute.
void noteAttributeNames(std::vector<std::string> names)
    Supplies the attribute names that declare() checks against.
bool has(const std::string& name) const
    Whether a variable of this name exists.
double get(const std::string& name) const
    The current value.
void set(const std::string& name, double value, SimTime now)
    Closes the time integral at `now`, then changes the value. Throws on
    an undeclared name, so a typo cannot become a second variable.
void reset()
    Back to the t=0 condition. Configuration survives; run state does not.
void resetStatistics(SimTime now)
    Discards measurements collected so far and starts again from now,
    without touching what is held. This is warm-up removal at block level.
void updateIntegrals(SimTime now)
    Closes every variable's integral up to now.
double timeAverage(const std::string& name) const
    The time-weighted average since measurement began.
const std::vector<std::string>& names() const
    The variables, in declaration order.
std::size_t count() const
    How many entities have passed through.
```


Chapter 6

The Flowchart

The blocks entities move through, and the narrow window each one gets on the engine.

A model is a graph of blocks. Each does one job to an entity and decides where it goes next. A block reaches the engine only through NodeContext, which exposes what a block may do and nothing else: it cannot touch the future event list, the statistics or the clock.

NodeContext

class · Node.hpp

The narrow facade through which a block touches the engine. A node can create, route, schedule and destroy, and cannot reach the future event list, the statistics or the clock.

```
explicit NodeContext(SimulationSystem& sim) : m_sim(sim)
    Constructor.

SimTime now() const
    The current simulation time.

RandomStream& rng()
    The shared random stream.

Trace& trace()
    The trace writer.

void route(Entity* e, INode* to)
    Hands the entity to the next block. A null target means it leaves the
    system.

void scheduleReturn(SimTime at, Entity* e, INode* node)
    Asks to be called back at a given time with this entity -- how a Process
    schedules the end of a service without seeing the event list.

void scheduleRenegadeCheck(SimTime at, Entity* e, INode* node)
    Schedules a patience timer.

void scheduleNextArrival(SimTime at, INode* source)
    A source asking to produce again. A separate call because the event
    type differs.

void setEntityType(Entity* e, const std::string& type)
    Retypes an entity through the engine, so the per-type counts move
    with it.

EvalContext evaluationContext(const Entity* e)
    Builds the EvalContext an expression needs. The only route by which a
    block reaches the variables and the model state.

VariableStore& variables()
    The global variable store.

Entity* createEntity()
    Makes a new entity.

void registerArrival(Entity* e)
    Marks an entity as demand the system received. Only a Create block
    calls it; batch representatives and duplicates are manufactured, not
    arrivals.

void destroy(Entity* e)
    Destroys an entity.
```

INode

class · Node.hpp

A block in the flowchart. Does one job to an entity and decides where it goes next. Everything from Create to Dispose is one of these, which is why the engine has no special case for arrivals.

```
explicit INode(std::string name) : m_name(std::move(name))
    Constructor.

const std::string& name() const
    The name this object was constructed with.

INode* next() const
    The default onward block.

virtual void setNext(INode* n)
    Sets the block this one routes to by default.
```

```
virtual void enter(NodeContext& ctx, Entity* e) = 0
    An entity has arrived at this block. Does the block's job and routes
    onward -- possibly not immediately.

virtual void onScheduledEvent(NodeContext& ctx, Entity* e)
    A previously scheduled return has come due, such as the end of a
    service or a delay.

virtual void onRenegadeTimeout(NodeContext& , Entity* /*e*/)
    A patience timer has expired. Timers are never cancelled, so this
    checks whether the entity is still waiting and ignores the event if not.

virtual void reset()
    Back to the t=0 condition. Configuration survives; run state does not.

virtual void resetStatistics(SimTime )
    Discards measurements collected so far and starts again from now,
    without touching what is held. This is warm-up removal at block level.

virtual std::string describe() const = 0
    A human-readable one-line description, used in model dumps and
    traces.

virtual double loadPerVisit() const
    Mean resource-time consumed per entity passing through, divided by
    capacity. Used by the stability check.

virtual bool loadIsKnown() const
    False when the mean cannot be computed in advance, which is not the
    same as no load.
```

DelayNode

class · Nodes.hpp · inherits public INode

Holds an entity for a duration and passes it on. No resource, no queue, no contention -- transport, cooling, walking between rooms.

```
DelayNode(std::string name, std::unique_ptr<IDistribution>
duration)
    Constructor.

DelayNode(std::string name, ExpressionPtr duration)
    Constructor.

const IExpression& durationExpression() const
    The expression giving the hold time.

IExpression& durationExpression()
    The expression giving the hold time.

void enter(NodeContext& ctx, Entity* e) override
    An entity has arrived at this block. Does the block's job and routes
    onward -- possibly not immediately.

void onScheduledEvent(NodeContext& ctx, Entity* e) override
    A previously scheduled return has come due, such as the end of a
    service or a delay.

void reset() override
    Back to the t=0 condition. Configuration survives; run state does not.

void resetStatistics(SimTime now) override
    Discards measurements collected so far and starts again from now,
    without touching what is held. This is warm-up removal at block level.

std::string describe() const override
    A human-readable one-line description, used in model dumps and
    traces.

const Statistics& stats() const
    The statistics accumulator for this block.

int inTransit() const
    The in transit.
```

AssignTarget

enum class · Nodes.hpp

What an Assign writes: an entity attribute, a global variable, or the entity's type.

Values Attribute, Variable, EntityType

AssignNode

class · Nodes.hpp · inherits public INode

Sets one or more values on the entity passing through, instantly. Fields run in order, so a later field sees what an earlier one wrote.

explicit AssignNode(std::string name)
Constructor.

AssignNode& set(const std::string& attribute, std::unique_ptr<IDistribution> value)
Adds a field to this block. Fields run in the order they were added.

AssignNode& set(AssignTarget target, const std::string& name, ExpressionPtr value)
Adds a field to this block. Fields run in the order they were added.

const std::vector<Rule>& rules() const
The fields this block writes, in order.

void enter(NodeContext& ctx, Entity* e) override
An entity has arrived at this block. Does the block's job and routes onward -- possibly not immediately.

void reset() override
Back to the t=0 condition. Configuration survives; run state does not.

void resetStatistics(SimTime now) override
Discards measurements collected so far and starts again from now, without touching what is held. This is warm-up removal at block level.

std::string describe() const override
A human-readable one-line description, used in model dumps and traces.

long long count() const
How many entities have passed through.

DecideNode

class · Nodes.hpp · inherits public INode

Branches. Either by chance, from a single draw walked against cumulative probabilities, or by condition, first match wins. A block is all-chance or all-condition, never mixed.

DecideNode(std::string name, double probability)
Constructor.

DecideNode(std::string name, Condition condition)
Constructor.

DecideNode(std::string name, ExpressionPtr condition)
Constructor.

explicit DecideNode(std::string name, bool byChance)
Constructor.

DecideNode& addBranch(double probability, INode* target)
Adds a branch, in order. Order is meaningful for conditions: first match wins.

DecideNode& addBranch(Condition condition, INode* target)
Adds a branch, in order. Order is meaningful for conditions: first match wins.

DecideNode& addBranch(ExpressionPtr condition, INode* target)
Adds a branch, in order. Order is meaningful for conditions: first match wins.

void setTrueBranch(INode* n)
Sets the target of the first branch -- the two-way spelling.

INode* trueBranch() const
The true branch.

void enter(NodeContext& ctx, Entity* e) override
An entity has arrived at this block. Does the block's job and routes onward -- possibly not immediately.

void reset() override
Back to the t=0 condition. Configuration survives; run state does not.

void resetStatistics(SimTime now) override
Discards measurements collected so far and starts again from now, without touching what is held. This is warm-up removal at block level.

std::string describe() const override
A human-readable one-line description, used in model dumps and traces.

bool isByChance() const
Whether this block branches by chance rather than by condition.

double probability() const
The probability.

const std::vector<Branch>& branches() const
The branches, in evaluation order.

long long fellThrough() const
How many matched no branch.

long long tookTrue() const
How many entities took the first branch.

long long tookFalse() const
How many took any other branch or fell through.

BatchNode

class · Nodes.hpp · inherits public INode

Accumulates entities and sends one representative onward carrying them. Permanent batches consume their members; temporary ones can be split again later. Can group by matching or by distinct attribute values.

BatchNode(std::string name, std::size_t size, bool permanent = false)
Constructor.

void enter(NodeContext& ctx, Entity* e) override
An entity has arrived at this block. Does the block's job and routes onward -- possibly not immediately.

void reset() override
Back to the t=0 condition. Configuration survives; run state does not.

void resetStatistics(SimTime now) override
Discards measurements collected so far and starts again from now, without touching what is held. This is warm-up removal at block level.

std::string describe() const override
A human-readable one-line description, used in model dumps and traces.

std::size_t size() const
The size.

bool isPermanent() const
Whether permanent.

long long batchesFormed() const
The batches formed.

std::size_t waitingForBatch() const
The waiting for batch.

std::size_t maxQueueLength() const
The max queue length.

const Statistics& queueStats() const
The queue stats.

const Statistics& stats() const
The statistics accumulator for this block.

Rule rule() const
The rule.

SeparateNode

class · Nodes.hpp · inherits public INode

Splits a temporary batch back into its members, or duplicates an entity. Has two exits -- the original and the duplicate -- which is a distinction easy to lose when reading an Arena model.

void setDuplicateExit(INode* n)
Sets the duplicate exit.

INode* duplicateExit() const
The duplicate exit.

void setPercentToDuplicates(double pct)
Sets the percent to duplicates.

double percentToDuplicates() const
The percent to duplicates.

explicit SeparateNode(std::string name)
Constructor.

SeparateNode(std::string name, int copies)
Constructor.

void enter(NodeContext& ctx, Entity* e) override
An entity has arrived at this block. Does the block's job and routes onward -- possibly not immediately.

void reset() override

Back to the t=0 condition. Configuration survives; run state does not.

void resetStatistics(SimTime now) override

Discards measurements collected so far and starts again from now, without touching what is held. This is warm-up removal at block level.

std::string describe() const override

A human-readable one-line description, used in model dumps and traces.

long long processed() const

The processed.

RecordNode

class · Nodes.hpp · inherits public INode

Tallies something as entities pass: a count, an attribute value, or time in system.

explicit RecordNode(std::string name)

Constructor.

RecordNode(std::string name, std::string attribute)

Constructor.

static std::unique_ptr<RecordNode> timeInSystem(std::string name)

Builds a Record block that tallies each entity's total time in the system.

void enter(NodeContext& ctx, Entity* e) override

An entity has arrived at this block. Does the block's job and routes onward -- possibly not immediately.

void reset() override

Back to the t=0 condition. Configuration survives; run state does not.

void resetStatistics(SimTime now) override

Discards measurements collected so far and starts again from now, without touching what is held. This is warm-up removal at block level.

std::string describe() const override

A human-readable one-line description, used in model dumps and traces.

long long count() const

How many entities have passed through.

double total() const

The total.

double average() const

The average.

double minimum() const

The minimum.

double maximum() const

The maximum.

DisposeNode

class · Nodes.hpp · inherits public INode

The exit. Records the entity's departure and destroys it.

explicit DisposeNode(std::string name)

Constructor.

void enter(NodeContext& ctx, Entity* e) override

An entity has arrived at this block. Does the block's job and routes onward -- possibly not immediately.

void setNext(INode*) override

Sets the next.

void reset() override

Back to the t=0 condition. Configuration survives; run state does not.

void resetStatistics(SimTime now) override

Discards measurements collected so far and starts again from now, without touching what is held. This is warm-up removal at block level.

std::string describe() const override

A human-readable one-line description, used in model dumps and traces.

long long count() const

How many entities have passed through.

Station

class · Station.hpp · inherits public INode, public IResourceUser

The Process block: seize a resource, hold the entity for a service time, release. Also the only block that queues, balks and reneges. Implements both INode and IResourceUser.

void enter(NodeContext& ctx, Entity* e) override

An entity has arrived at this block. Does the block's job and routes onward -- possibly not immediately.

void onScheduledEvent(NodeContext& ctx, Entity* e) override

A previously scheduled return has come due, such as the end of a service or a delay.

void onRenegadeTimeout(NodeContext& ctx, Entity* e) override

A patience timer has expired. Timers are never cancelled, so this checks whether the entity is still waiting and ignores the event if not.

void reset() override

Back to the t=0 condition. Configuration survives; run state does not.

void resetStatistics(SimTime now) override

Discards measurements collected so far and starts again from now, without touching what is held. This is warm-up removal at block level.

std::string describe() const override

A human-readable one-line description, used in model dumps and traces.

double loadPerVisit() const override

Mean resource-time consumed per entity passing through, divided by capacity. Used by the stability check.

bool hasWaiting() const override

Whether anything is queued here.

SimTime headOfLineSince() const override

When the longest-waiting entity here started waiting. A shared resource compares this across blocks.

void startFromQueue(NodeContext& ctx) override

Starts service for the next queued entity, if a unit is available.

void setServiceFromAttribute(const std::string& attributeName)

Takes the service time from a named attribute the entity already carries.

bool usesServiceAttribute() const

The uses service attribute.

const std::string& serviceAttributeName() const

The service attribute name.

void setRandomStream(RandomStream* rng)

Sets the random stream.

void setBalking(std::size_t queueLength, INode* balkTo)

Refuses to join a queue already this long, sending the entity elsewhere instead.

void setReneging(std::unique_ptr<IDistribution> patience, INode* renegeTo)

Gives a waiting entity a random patience, after which it gives up.

void setUnitsNeeded(int units)

Sets the units needed.

Resource& resource()

The resource this block seizes. May be shared with other blocks.

const Resource& resource() const

The resource this block seizes. May be shared with other blocks.

EntityQueue& queue()

The waiting line at this block.

const EntityQueue& queue() const

The waiting line at this block.

Statistics& stats()

The statistics accumulator for this block.

const Statistics& stats() const

The statistics accumulator for this block.

IExpression& serviceExpression()

The expression giving the service time.

const IExpression& serviceExpression() const

The expression giving the service time.

bool loadIsKnown() const override

False when the mean cannot be computed in advance, which is not the same as no load.

std::size_t stillWaiting() const

The still waiting.

```

long long balked() const
    The balked.
long long reneged() const
    The reneged.
int unitsHeld() const
    Units of the resource this block currently holds -- not the same as the
    resource's total busy count when it is shared.
int unitsNeeded() const
    The units needed.
SimTime drawService(NodeContext& ctx, const Entity& e)
    Evaluates the service-time expression for this entity.

```

CreateNode

class · Create.hpp · inherits public INode

An arrival source. Makes entities on its own schedule and reschedules itself. Because it is an ordinary INode, the engine has no separate arrival handler at all.

```

void enter(NodeContext& ctx, Entity* e) override
    An entity has arrived at this block. Does the block's job and routes
    onward -- possibly not immediately.
void onScheduledEvent(NodeContext& ctx, Entity* e) override
    A previously scheduled return has come due, such as the end of a
    service or a delay.
void reset() override
    Back to the t=0 condition. Configuration survives; run state does not.
void resetStatistics(SimTime now) override
    Discards measurements collected so far and starts again from now,
    without touching what is held. This is warm-up removal at block level.
std::string describe() const override
    A human-readable one-line description, used in model dumps and
    traces.
const std::string& entityType() const
    The entity type.
SimTime firstAt() const
    The first at.
long long created() const
    The created.
long long maxArrivals() const
    The max arrivals.
int entitiesPerArrival() const
    The entities per arrival.
IEExpression& interarrival()
    The expression giving the time until the next arrival. Evaluated with NO
    entity, because the entity does not exist yet.
const IEExpression& interarrival() const
    The expression giving the time until the next arrival. Evaluated with NO
    entity, because the entity does not exist yet.
bool exhausted() const
    Whether this source has produced its maximum number of arrivals.

```

Chapter 7

The Model

What is simulated, kept separate from the machinery that simulates it.

The engine takes a Model and runs it, and knows nothing about tellers or restaurants. Model owns the blocks, the resources and the variables, and validates the flowchart before a run, catching a routing loop as an error rather than as a program that never stops.

Nested types

ArrivalAttribute is a name and a distribution, stamped on every arrival. **VisitRatios** is how often an average entity reaches each block, with an exact flag that goes false when a condition-based Decide makes the split an output of the run rather than an input. **StabilityReport** is what the offered-load check could and could not judge.

Model

class · Model.hpp

What is simulated, as opposed to the machinery that simulates it. Owns the blocks, the resources and the variables, wires them into a flowchart, and validates the result before a run.

```
Model& arrivals(std::unique_ptr<IDistribution> d)
    The single-source shorthand: creates a source called Arrivals.

Model& arrivals(const std::string& interarrivalText)
    The single-source shorthand: creates a source called Arrivals.

Model& allowOverload(bool on = true)
    Permits a model whose queues grow without bound. Correct for a
    terminating run, meaningless for a steady-state study, and the report
    says so.

bool overloadAllowed() const
    Whether overload has been permitted.

void setInterarrival(std::unique_ptr<IDistribution> d)
    Sets the interarrival.

Model& variable(const std::string& name, double initialValue
= 0.0)
    Declares a global variable. Checks it does not collide with an attribute
    name.

VariableStore& variables()
    The variable store.

const VariableStore& variables() const
    The variable store.

Model& attribute(const std::string& name,
std::unique_ptr<IDistribution> d)
    Declares an attribute stamped on every arrival, drawn from a
    distribution.

void assignOnArrival(const std::string& n,
std::unique_ptr<IDistribution> d)
    Alias for attribute(): stamps a drawn value on every arriving entity.

const std::vector<ArrivalAttribute>& arrivalAttributes()
const
    The attributes stamped on every arrival.

Model& resource(const std::string& name, int capacity)
    Declares a shared resource that several blocks may seize.

Resource* resourceNamed(const std::string& name)
    Looks a resource up by name.

const Resource* resourceNamed(const std::string& name) const
    Looks a resource up by name.

Model& decideNWayByChance(const std::string& name)
    Starts an N-way Decide branching by chance.

Model& decideNWayByCondition(const std::string& name)
    Starts an N-way Decide branching by condition.

Model& branch(const std::string& decideName, double
probability, const std::string& to)
    Adds a branch to an N-way Decide, in order.

Model& delay(const std::string& name,
std::unique_ptr<IDistribution> duration)
```

Adds a Delay block.

```
Model& delay(const std::string& name, const std::string&
durationText)
```

Adds a Delay block.

```
Model& assignEntityType(const std::string& block, const
std::string& typeText)
```

Restamps the entity's type from an expression.

```
Model& decideByChance(const std::string& name, double
probabilityTrue)
```

Adds a two-way Decide taking the true branch with a fixed probability.

```
Model& decideByCondition(const std::string& name,
DecideNode::Condition condition)
```

Adds a two-way Decide on a predicate or expression.

```
Model& decideByCondition(const std::string& name,
ExpressionPtr condition)
```

Adds a two-way Decide on a predicate or expression.

```
Model& decideWhen(const std::string& name, const
std::string& conditionText)
```

Adds a two-way Decide whose condition is written as text.

```
Model& batch(const std::string& name, std::size_t size, bool
permanent = false)
```

Accumulates entities into groups of a given size.

```
Model& separate(const std::string& name)
```

Splits a temporary batch back into its members.

```
Model& duplicate(const std::string& name, int copies)
```

Copies an entity a given number of times.

```
Model& record(const std::string& name)
```

Counts entities passing a point.

```
Model& recordAttribute(const std::string& name, const
std::string& attributeName)
```

Tallies an attribute's value as entities pass.

```
Model& recordTimeInSystem(const std::string& name)
```

Tallies time in system as entities pass.

```
Model& dispose(const std::string& name)
```

Adds a named exit.

```
Model& route(const std::string& from, const std::string& to)
```

Connects one block's default exit to another block.

```
Model& routeTrue(const std::string& decideName, const
std::string& to)
```

Connects a Decide's true branch.

```
Model& routeDuplicate(const std::string& separateName, const
std::string& to)
```

Connects a Separate's duplicate exit. Its original exit is plain route().

```
Model& entryAt(const std::string& name)
```

Names the block arrivals enter at.

```
void connect(const std::string& f, const std::string& t)
```

Alias for route().

```
void setEntry(const std::string& n)
```

Alias for entryAt().

```
INode* node(const std::string& name)
```

Looks a block up by name.

```
const INode* node(const std::string& name) const
```

Looks a block up by name.

```
INode* entry() const
```

The block arrivals enter at.

```
template <typename T> T& nodeAs(const std::string& name)
```

Looks a block up by name and type, throwing if either is wrong rather than returning a null the caller will dereference later.

```
Station* station(const std::string& name)
```

Looks a Process block up by name.

```
const Station* station(const std::string& name) const
```

Looks a Process block up by name.

```
std::size_t stationCount() const
```

How many Process blocks the model has.

Station& stationAt(std::size_t i)
The i-th Process block. Process blocks are the only ones a utilisation table means anything for.

const Station& stationAt(std::size_t i) const
The i-th Process block. Process blocks are the only ones a utilisation table means anything for.

std::size_t sourceCount() const
How many arrival sources the model has.

CreateNode& sourceAt(std::size_t i)
The i-th arrival source.

const CreateNode& sourceAt(std::size_t i) const
The i-th arrival source.

std::size_t nodeCount() const
How many blocks the model has in total.

Inode& nodeAt(std::size_t i)
The i-th block, of any kind.

const Inode& nodeAt(std::size_t i) const
The i-th block, of any kind.

StabilityReport stability() const
What the offered-load check could and could not verify.

bool arrivalRateIsKnown() const
False when any source's interarrival mean cannot be computed in advance, which makes every station's load unknowable too.

std::vector<Diagnostic> checkExpressions() const
Validates every expression in the model against the declared names, returning diagnostics rather than throwing.

VisitRatios visitRatios() const
How many times an average entity passes through each block.

double offeredLoad(const Station& s) const
The offered load at one Process, summed over every arrival stream and weighted by visits.

void wireSources()
Connects the shorthand source to the entry block, so the order a model is described in does not matter.

void reset()
Back to the t=0 condition. Configuration survives; run state does not.

void validate() const
Checks the whole model and throws on the first problem. Catches routing loops before they become a hang.

std::string describe() const
The whole model as text, block by block.

Chapter 8

Running a Simulation

The engine, the stopping rules, and the trace that makes a run checkable by hand.

SimulationSystem owns the clock, the event list, the entities, the random streams and the model. It also implements IModelState, which is how an expression reads the running system without the expression layer depending on the model.

SimulationSystem

class · SimulationSystem.hpp · inherits public IModelState

The engine. Owns the clock, the future event list, the entities, the random streams and the model, and runs the loop that drives them. Also implements IModelState so expressions can read the running system.

```
explicit SimulationSystem(unsigned seed = 12345u)
    Constructor.

double queueLength(const std::string& blockName) const
    override
    How many entities wait at a named block. Part of IModelState, so
    expressions can read it. Throws on an unknown name rather than
    reading as an empty queue.

double resourceBusy(const std::string& name) const override
    Units of a named resource currently seized. A Process with a private
    resource is addressable by the block's name.

double resourceCapacity(const std::string& name) const
    override
    A named resource's capacity.

double numberInSystem() const override
    How many entities are anywhere in the system.

SimTime now() const override
    The current simulation time.

double variableAverage(const std::string& name) const
    The time-weighted average of a global variable.

void reportStability(std::ostream& os) const
    Prints the blocks the offered-load check could not verify. Prints nothing
    when everything was checked.

Model& model()
    The model being simulated.

const Model& model() const
    The model being simulated.

void setTermination(std::unique_ptr<ITerminationRule> rule)
    Sets an arbitrary stopping rule.

void setWarmUp(SimTime t)
    Discards statistics collected before this time.

void setObservationInterval(SimTime dt)
    Samples number-in-system on a fixed grid, for Welch plots.

SimulationSystem& useSeparateStreams(bool on = true)
    Gives each named role its own random stream, which is what makes
    two model variants comparable.

SimulationSystem& useAntithetic(bool on = true)
    Turns on antithetic variates.

bool isAntithetic() const
    Whether antithetic.

const std::vector<double>& observations() const
    The observations.

SimTime measuredTime() const
    The measured time.

SimTime warmUpEnd() const
    The warm up end.

bool enableTrace(const std::string& path, TraceLevel level,
bool markdown = true)
    Starts writing an event log to a file.

const Clock& clock() const
    The simulation clock.

const Statistics& statistics() const
```

The statistics.

```
const SystemState& state() const
    The state.

const FutureEventList& fel() const
    The fel.

const RandomStream& randomStream() const
    The random stream.

std::size_t liveEntityCount() const
    The live entity count.

Entity* createEntity()
    Makes an entity and registers it in the entity table. Called through
    NodeContext, never directly by a block.

void initialise()
    Validates the model, resets everything holding run state, and
    schedules the first arrival. Must be called before run().

void run()
    The event loop: pop the earliest event, close the integrals for the
    interval that just ended, move the clock, then dispatch. That order is
    not negotiable.

void report() const
    Prints the run's results.

void reportArenaStyle() const
    Prints the same numbers in Arena's report layout.

SimulationSystem& execute()
    initialise() and run() in one call, for the common case.

RunResults results() const
    The run's numbers, derived once.

const std::map<std::string, TypeStats>& byType() const
    Per-entity-type counts and times.

SimulationSystem& stopWhen(std::unique_ptr<ITerminationRule>
rule)
    Sets an arbitrary stopping rule, returning *this so it chains.

SimulationSystem& stopAt(SimTime t)
    Stops the run at a given time.

SimulationSystem& stopAfter(int entities)
    Stops after a given number of entities have left.

SimulationSystem& warmUpFor(SimTime t)
    Discards statistics collected before this time, returning *this so it
    chains.
```

StationResults

struct · SimulationSystem.hpp

One block's numbers after a run: served, utilisation, average wait, queue length.

RunResults

struct · SimulationSystem.hpp

The whole run's numbers, derived once so that printing is only a view of them.

```
const StationResults& station(const std::string& name) const
    One block's results, by name.
```

ITerminationRule

class · TerminationRule.hpp

The interface a stopping condition implements. Asked after every event whether the run should end.

```
virtual bool isMet(const SimulationSystem& sim) const = 0
    Whether this stopping condition now holds.

virtual std::string describe() const = 0
```

A human-readable one-line description, used in model dumps and traces.

TimeLimit

class · TerminationRule.hpp · inherits public ITerminationRule

Stop when the clock reaches a given time.

explicit TimeLimit(SimTime maxTime)
Constructor.

bool isMet(const SimulationSystem& sim) const override
Whether this stopping condition now holds.

std::string describe() const override
A human-readable one-line description, used in model dumps and traces.

SimTime maxTime() const
The max time.

EntityLimit

class · TerminationRule.hpp · inherits public ITerminationRule

Stop after a given number of entities have left the system.

explicit EntityLimit(int maxEntities)
Constructor.

bool isMet(const SimulationSystem& sim) const override
Whether this stopping condition now holds.

std::string describe() const override
A human-readable one-line description, used in model dumps and traces.

DrainedRule

class · TerminationRule.hpp · inherits public ITerminationRule

Stop when nothing is left in the system. Fires at the first moment the system is empty, which may be a transient gap rather than the end of the work.

bool isMet(const SimulationSystem& sim) const override
Whether this stopping condition now holds.

std::string describe() const override
A human-readable one-line description, used in model dumps and traces.

AnyOf

class · TerminationRule.hpp · inherits public ITerminationRule

Stops when any composed rule says stop.

AnyOf& add(std::unique_ptr<ITerminationRule> rule)
Adds a rule; the composite stops when any of them does.

bool isMet(const SimulationSystem& sim) const override
Whether this stopping condition now holds.

std::string describe() const override
A human-readable one-line description, used in model dumps and traces.

bool empty() const
Whether any rules have been added. An AnyOf with none never stops the run.

TraceLevel

enum class · Trace.hpp

How much detail the trace records, from off to every event.

Values Off, Events, Detailed

Trace

class · Trace.hpp

Writes an event-by-event log, optionally as a markdown table. Traces diff, which is what makes a behaviour change show up as a diff rather than as a slightly-off average.

~Trace()

Destructor.

bool open(const std::string& path, TraceLevel level, bool markdown = true)

Starts writing a trace to a file.

void close()

Closes the trace file. Called by the destructor, so this is only for closing early.

bool isOn() const

Whether tracing is enabled. Callers check this before building a message.

TraceLevel level() const

The level.

long long lineCount() const

How many lines have been written, so a test can assert a trace was produced.

void note(const std::string& text)

Writes a free-text line into the trace, outside the event table.

Chapter 9

Experiments

Why one run is one sample, and what to do about it.

A single run is one observation of a random variable. Quoting it to four decimals implies a precision that does not exist. Experiment runs the model many times with independent seeds and reports an interval.

Nested types

Estimate is a point estimate with its confidence interval.

Comparison is two configurations and the interval on their difference, which is what actually answers whether a change did anything.

ReplicationResult

struct · Experiment.hpp

One replication's summary figures, so a set of runs can be treated as a sample.

Summary

class · Experiment.hpp

The statistics of a sample of replications: mean, standard deviation, standard error and a 95% half-width from Student's t.

```
static double mean(const std::vector<double>& xs)
    The theoretical mean, used by the stability check before the run.

static double stdDev(const std::vector<double>& xs)
    Sample standard deviation, with the n-1 denominator.

static double standardError(const std::vector<double>& xs)
    Standard deviation divided by the square root of the sample size.

static double halfWidth95(const std::vector<double>& xs)
    Half the width of a 95% confidence interval, from Student's t.

static double tCritical95(int degreesOfFreedom)
    The two-sided 95% critical value of Student's t.
```

Experiment

class · Experiment.hpp

Runs a model many times with independent seeds and reports an interval rather than a number. Also implements Welch's method for finding the warm-up period.

```
Experiment(std::string name, Builder build)
    Constructor.

Experiment& replications(int n)
    How many independent runs to make.

Experiment& baseSeed(unsigned s)
    The seed the per-replication seeds are derived from.

Experiment& warmUp(SimTime t)
    The warm-up period applied to every replication.

Experiment& observeEvery(SimTime dt)
    The observation grid used for Welch's method.

Experiment& separateStreams(bool on = true)
    Turns on per-role random streams for every replication.

Experiment& antitheticPairs(bool on = true)
    Pairs each replication with its antithetic twin.

void run()
    Runs every replication and collects the results.

const std::vector<ReplicationResult>& results() const
    The results.

std::vector<double> column(double ReplicationResult::*
field) const
    Pulls one field out of every replication result as a plain vector, so any of
    them can be summarised the same way.

std::vector<double> waits() const
```

The waits.

```
std::vector<double> timesInSystem() const
    The times in system.

std::vector<double> queueLengths() const
    The queue lengths.

std::vector<double> utilisations() const
    The utilisations.

static Estimate estimate(const std::vector<double>& xs)
    A point estimate with its interval, from a sample.

std::vector<double> welchAverages(int window) const
    The Welch moving average across replications, for finding the
    transient.

SimTime suggestWarmUp(int window = 0, double maxFraction =
0.5) const
    A warm-up length suggested by MSER. A starting point, not an
    answer.

bool writeWelchSeries(const std::string& path, int window)
const
    Writes the Welch series to a CSV so it can be plotted.

void report() const
    Prints each quantity as a mean with a 95% confidence interval.
```

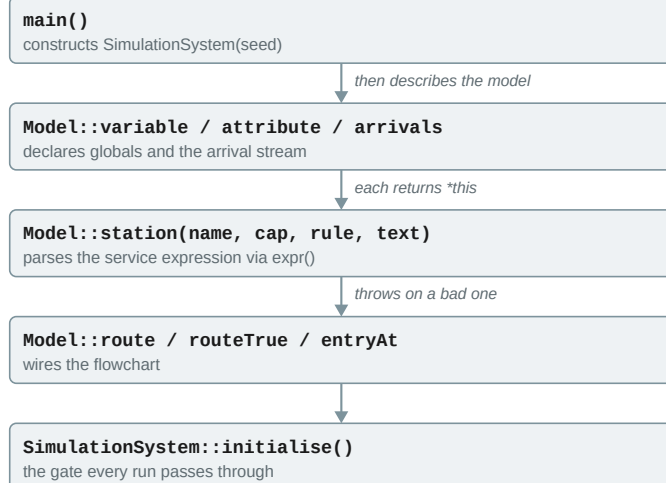

Chapter 10

Call Flows

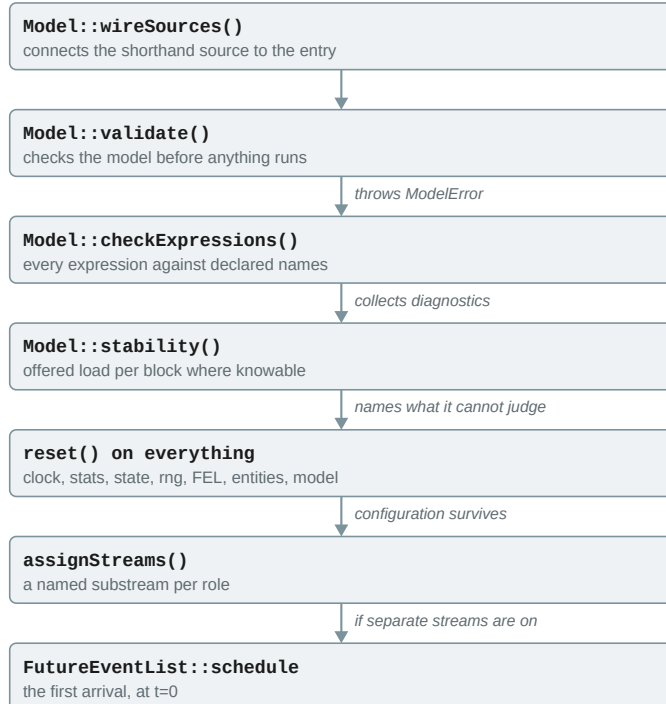
What calls what, in order, when a model is built and run.

These traces follow examples/16_expressions.cpp, which exercises every part of the engine: text expressions, a condition reading a queue, a global variable, and a Decide by chance.

1 - building and starting

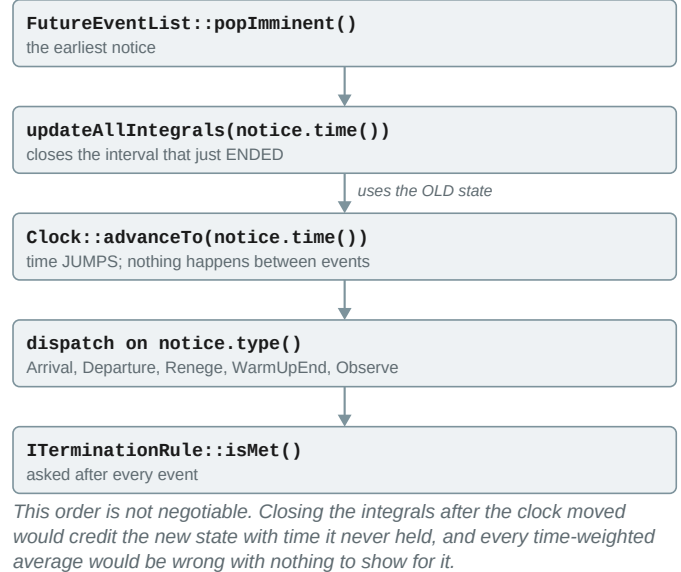


2 - initialise()

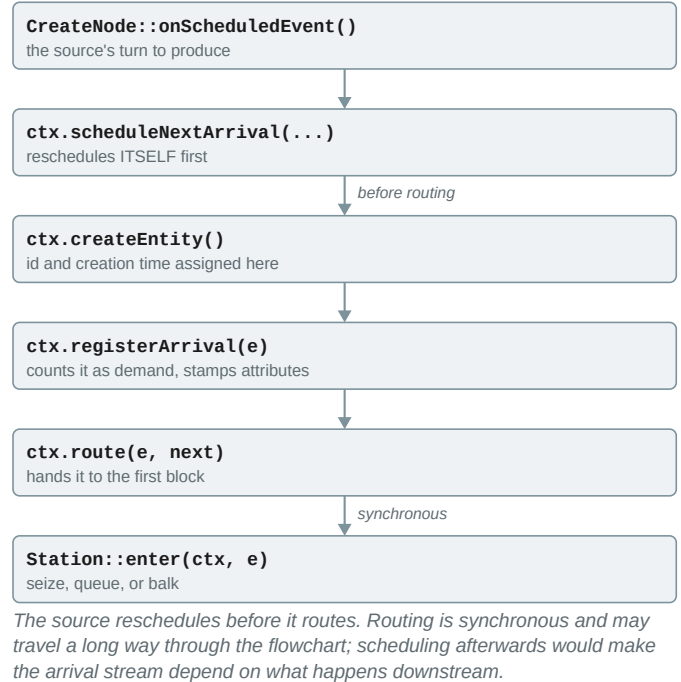


Order matters here: the model is validated BEFORE run state is reset, so a broken model fails without having touched anything.

3 - the run loop



4 - an arrival



5 - service, and a freed unit



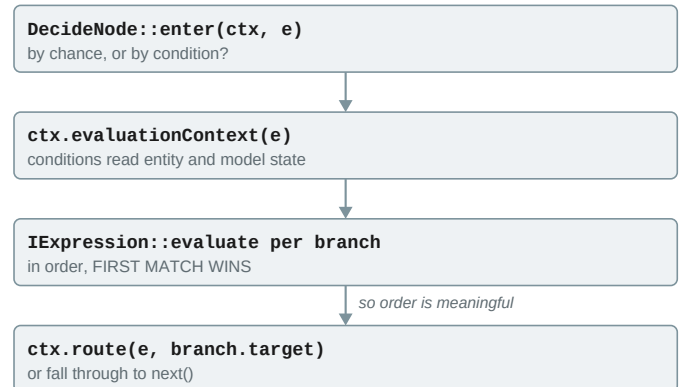
The freed unit is offered through the resource rather than handed back to this block's own queue, because with a shared resource somebody at another block may have been waiting longer.

6 - evaluating an expression



The expression layer declares *IModelState* and *SimulationSystem* implements it, so the dependency points downward and the expression layer never includes *Model.hpp*.

7 - a Decide by condition



A by-chance Decide takes ONE draw and walks it against the cumulative probabilities. Drawing once per branch would burn several random numbers and would not give the branch probabilities you asked for.

Chapter 11

Free Functions

The factories and helpers that are not members of anything.

Distribution factories, from Build.hpp

Thin wrappers so a model reads as a description rather than as memory management.

Function	Builds
exponential(mean)	Exponential. Takes a MEAN, not a rate.
constant(v)	Constant. Consumes nothing from the stream.
uniform(a, b)	Uniform on a closed interval.
triangular(a, mode, b)	Triangular.
normal(mean, sd, truncate)	Normal, truncated at zero by default.
lognormal(logMean, logSd)	Lognormal from log-space parameters.
lognormalFrom(mean, sd)	Lognormal from observed mean and sd.
weibull(scale, shape)	Weibull.
erlang(totalMean, phases)	Erlang with the mean you want.
discrete(values, probs)	Discrete, individual probabilities.
empirical(observations)	Empirical, sampled from data.
poisson(mean)	Poisson counts.
fixedTimes(values, repeat)	Deterministic. The basis of hand checks.

Termination rules, from Build.hpp

Function	Stops when
timeLimit(t)	the clock reaches t.
entityLimit(n)	n entities have left.
whenDrained()	the system is empty, which may be a transient gap.
anyOf(a, b, ...)	any composed rule fires.

The expression layer

Function	Does
tokenise(text)	Text to tokens with spans. Never throws.
parseExpression(text)	Tokens to a tree plus diagnostics. Never throws.
expr(text)	The same, but throws. For C++ callers.
isBuiltinFunction(name)	Whether a name is in the call table.
isNumber / isText	Which half of a Value is present.
asNumber / asText	Reads a Value, throwing rather than coercing.
truthy(v)	Non-zero is true; text throws.
formatValue(v)	A Value as text, whole numbers without a point.
hasErrors(diags)	Whether any diagnostic is an error.
formatDiagnostics(diags)	Diagnostics as text, one per line.
describe(TokenKind)	Names a token kind, for messages.
spelling(op)	An operator's symbol.

Grammar functions

Usable inside any expression field. Distributions: EXPO, CONS, UNIF, TRIA, NORM, LOGN, WEIB, ERLA, POIS, DISC. Model state: NQ(block), NR(name), MR(name), WIP(), and the constant TNOW. Maths: MIN, MAX, ABS, ROUND, TRUNC, SQRT, LN, EXP, MOD.

Three traps. EXPO is the exponential distribution while EXP is e to the x, which is Arena's collision kept deliberately. DISC takes CUMULATIVE probabilities as Arena writes them, while the engine's own discrete() takes individual ones. LOGN takes the mean and standard deviation of the variable, not of its logarithm.